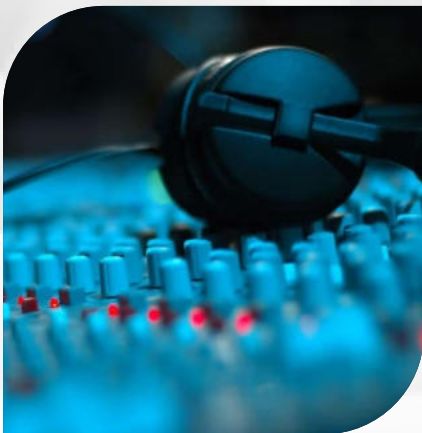


UNIT 11

SOUND



SOUND WAVES

Sound waves are vibrations of air particles that travel through a medium, such as air, and can be detected by our ears. These waves carry sound energy and enable us to hear sounds in our surroundings.

PRODUCTION OF SOUND BY VIBRATING SOURCES

When an object vibrates, it causes the air particles around it to vibrate as well. These vibrations create pressure variations in the air, which then propagate as sound waves. The frequency of the vibrations determines the pitch of the sound, while the amplitude of the vibrations affects the sound loudness.

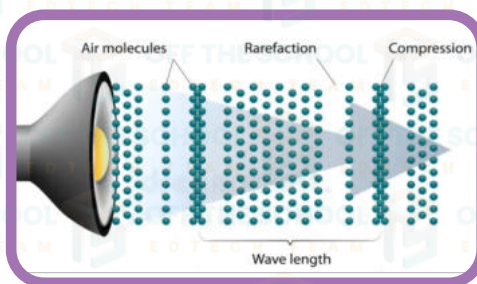
LONGITUDINAL NATURE OF SOUND WAVES

Sound waves are longitudinal waves, which means that the particles in the medium vibrate parallel to the direction of wave propagation.

1. Compressions:

Compressions are regions in a sound wave where air particles are densely packed together, leading to higher air pressure.

2. Rarefactions:



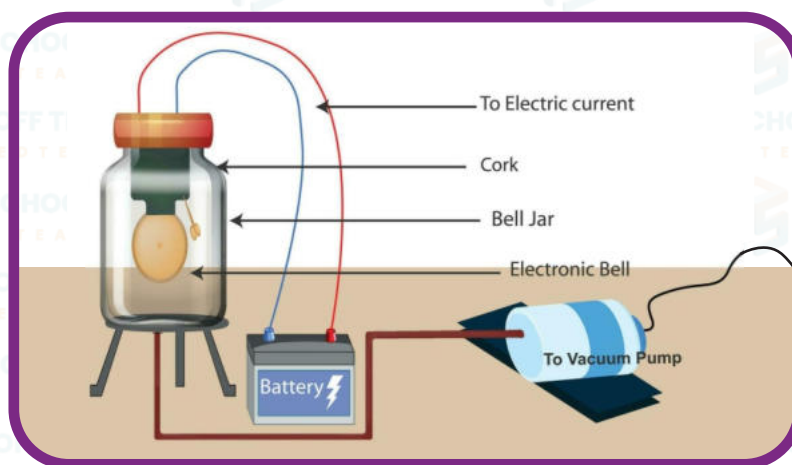
Rarefactions are regions in a sound wave where air particles are spread apart, resulting in lower air pressure compared to the surrounding areas.

ELECTRIC BELL JAR EXPERIMENT

The electric bell jar experiment was conducted to demonstrate that sound cannot travel through a vacuum, as sound requires a medium to propagate, such as air, water, or solids.

EXPERIMENT

1. Set up an electric bell and a vacuum bell jar apparatus. The electric bell consists of a vibrating clapper that strikes the bell when an electric current pass through it.
2. Connect the electric bell to a power source and ensure it is working correctly, producing audible sound in the surrounding air.
3. Place the electric bell apparatus inside the vacuum bell jar and seal it airtight.
4. Gradually pump out the air from the vacuum bell jar using a vacuum pump until a near-vacuum condition is achieved.
5. Observe that as the air is removed from the jar and the pressure decreases, the sound produced by the electric bell becomes fainter and eventually cannot be heard at all.
6. This observation confirms that sound requires a medium, such as air, to travel, as in the absence of air (vacuum), the sound waves have no particles to propagate through, resulting in silence.



The electric bell jar experiment provides evidence that sound waves cannot travel through a vacuum, supporting the understanding that sound propagation relies on the presence of a medium for its transmission.

SPEED OF SOUND

Direct Method for the Determination of the Speed of Sound in Air:

To determine the speed of sound in air, there are two primary methods:

METHOD 1: MEASURING SOUND BETWEEN TWO POINTS

In this method, an observer measures the time taken for sound to travel between two points at a known distance apart. By dividing the distance by the time taken, the speed of sound in air can be calculated.

METHOD 2: USING ECHOES

In this method, a loud sound is produced at a specific location, and the time it takes for the echo to return after reflecting off a distant surface is measured. By knowing the distance between the sound source and the reflecting surface, the speed of sound can be calculated using the time taken for the echo to travel back and forth.

SPEED OF SOUND IN SOLIDS, LIQUIDS, AND GASES

The speed of sound varies depending on the medium through which it travels. It is generally faster in solids, followed by liquids, and slowest in gases. The speed of sound in a given medium is influenced by its elasticity and density.

Speed of Sound in Solids: In solids, the particles are closely packed, and the intermolecular forces are strong, leading to a high elasticity of the medium. The speed of sound in a solid is given by the formula:

$$v_s = E/\rho$$

Where:

v_s = Speed of sound in the solid

E = Young's modulus (a measure of the solid's elasticity)

ρ = Density of the solid

SPEED OF SOUND IN LIQUIDS

Liquids have moderate particle density and elasticity compared to solids. The speed of sound in a liquid can be approximated using the formula:

$$v_l = \sqrt{\frac{K}{\rho}}$$

Where:

v_l = Speed of sound in the liquid

K = Bulk modulus (a measure of the liquid's compressibility)

ρ = Density of the liquid

SPEED OF SOUND IN GASES

Gases have low particle density and elasticity compared to solids and liquids. The speed of sound in a gas is given by:

$$v_g = \sqrt{\frac{\gamma P}{\rho}}$$

Where:

v_g = Speed of sound in the gas

γ = Adiabatic index or ratio of specific heat capacities $\frac{c_p}{c_v}$ for the gas

P = Pressure of the gas

ρ = Density of the gas

In general, the speed of sound is faster in denser and more elastic media, and it is slower in less dense and less elastic media. Understanding the variations in the speed of sound in different materials is crucial in various scientific and engineering applications.

FACTORS THAT AFFECT THE SPEED OF SOUND

1. Medium's Elasticity:

The speed of sound is directly proportional to the elasticity of the medium. In more elastic materials like solids, sound travels faster compared to less elastic mediums like gases.

2. Medium's Density:

The speed of sound is inversely proportional to the density of the medium. In denser materials, such as solids, sound waves propagate faster, while in less dense mediums like gases, sound travels at a slower pace.

3. Temperature of the Medium:

The speed of sound increases with a rise in temperature for gases, liquids, and solids. Warmer temperatures cause particles to vibrate more rapidly, facilitating faster sound transmission through medium.

EFFECT OF TEMPERATURE

The speed of sound in gases, liquids, and solids is influenced by the temperature of the medium. As the temperature increases, the speed of sound also increases due to the changes in the molecular motion within the medium.

FORMULA FOR THE EFFECT OF TEMPERATURE ON THE SPEED OF SOUND IN GASES

For an ideal gas, the relationship between the speed of sound v_g and temperature T is given by:

$$v_g = \sqrt{\frac{\gamma RT}{M}}$$

Where:

v_g = Speed of sound in the gas

γ = Adiabatic index or ratio of specific heat capacities $\frac{c_p}{c_v}$ for the gas

R = Gas constant

T = Temperature of the gas in Kelvin (K)

M = Molar mass of the gas

This equation shows that the speed of sound in a gas is directly proportional to the square root of the temperature. As the temperature increases, the molecular motion and kinetic energy of the gas molecules increase, resulting in a higher speed of sound. Apologies for the oversight. When the medium is air (which is a mixture of gases), the formula to find the effect of temperature on the speed of sound is very similar to the one for ideal gases, with some adjustments to account for the specific properties of air:

FORMULA FOR THE EFFECT OF TEMPERATURE ON THE SPEED OF SOUND IN AIR

The speed of sound v_a in air is given by the equation:

$$v_a = 331.5 \sqrt{1 + \frac{T}{273.15}}$$

Where:

v_g = Speed of sound in air (in meters per second, m/s)

T = Temperature of the air in degrees Celsius ($^{\circ}\text{C}$)

In this formula, 331.5 m/s is the speed of sound in air at 0°C (273.15 K). The term $\frac{T}{273.15}$ accounts for the change in speed due to temperature variation from 0°C . As the temperature increases, the speed of sound in air also increases accordingly.

EFFECT OF HUMIDITY

Humidity, which refers to the amount of moisture present in the air, affects the speed of sound. Higher humidity levels lead to a decrease in the speed of sound in air due to the presence of water vapor molecules, which influence the air's density and compressibility.

SEEING SOUNDS

Humidity, which refers to the amount of moisture present in the air, affects the speed of sound. Higher humidity levels lead to a decrease in the speed of sound in air due to the presence of water vapor molecules, which influence the air's density and compressibility.

1. Quality:

Quality refers to the timbre or tonal color of a sound, which allows us to distinguish between different sources of sound, such as musical instruments or voices.

2. Loudness:

Loudness corresponds to the amplitude or intensity of a sound wave and determines how soft or loud a sound appears to our ears.

3. Pitch:

Pitch is related to the frequency of a sound wave and defines how high or low a sound is perceived.

4. Sound Intensity:

Sound intensity measures the amount of sound energy that passes through a unit area and affects how "strong" a sound is perceived by our ears.

NOISE POLLUTION

Noise pollution refers to the excessive and harmful levels of noise in the environment that can disrupt natural soundscape, cause health issues, and interfere with communication and daily activities.

NOISE LEVELS IN DECIBELS FOR VARIOUS NOISES

Noises	Noise Level (in decibels, dB)
Normal Conversation	60 – 70 dB
City Traffic	70 – 85 dB
Vacuum Cleaner	70 – 80 dB
Loud Music (at concerts)	100 – 120 dB
Chainsaw	100 – 110 dB
Jet Engine (at take-off)	120 – 140 dB

REFLECTION OF SOUND OR AN ECHO

Reflection of sound, also known as an echo, occurs when sound waves encounter a surface and bounce back towards the source. The phenomenon of an echo can be heard when the reflected sound reaches the listener's ears after a noticeable time delay, which depends on the distance between the sound source and the reflecting surface, the speed of sound in the medium, and the time taken for the sound to travel to the surface and back.

FORMULA FOR CALCULATING DISTANCE OR SPEED FROM AN ECHO

The distance (d) between the sound source and the reflecting surface can be calculated using the formula:

$$d = \frac{v}{2}t$$

Where:

v = Speed of sound in air

t = Time taken for the sound to travel to the surface and back

Similarly, the speed of sound (v) can be calculated when both the distance and time are known:

$$v = \frac{2d}{t}$$

ULTRASOUND

Ultrasound refers to sound waves with frequencies higher than the upper limit of the human audible range, typically above 20,000 hertz (Hz). It is widely used in medical imaging, industrial applications, and scientific research due to its ability to penetrate materials and produce detailed images.

AUDIBLE FREQUENCY RANGE

The audible frequency range for humans typically ranges from 20 Hz to 20,000 Hz. Sounds within this range can be detected by the human ear. Frequencies below this range are called infrasonic, while frequencies above this range are classified as ultrasonic or ultrasound.

INFRASONIC

Infrasonic refers to sound waves with frequencies below the lower limit of human audibility, typically below 20 Hz. Although humans cannot hear infrasonic sounds, some animals, such as elephants and whales, are sensitive to these low-frequency waves and use them for communication and navigation.

Species	Lower Limit of Audible Frequency (Hz)	Upper Limit of Audible Frequency (Hz)
Humans	20	20,000
Dogs	40	60,000
Cats	48	85,000
Bats	1,000	150,000
Dolphins	7,000	160,000
Elephants	14	12,000
Whales (Blue Whale)	10	20,000

APPLICATIONS OF ULTRASOUND TECHNIQUES IN INDUSTRY AND MEDICINE

Ultrasound techniques find diverse applications in both industry and medicine, harnessing the properties of high-frequency sound waves for various purposes.

1. Cleansing:

In the industry, ultrasound is utilized for cleaning delicate or intricate components by generating tiny bubbles that collapse and create pressure waves, effectively removing contaminants without causing damage.

2. Quality Control:

In industrial quality control, ultrasound is employed to detect defects or flaws in materials by analyzing the response of sound waves when passing through the materials, ensuring product integrity and safety.

3. Sound Navigation and Ranging (SONAR):

Derived from "Sound Navigation and Ranging," SONAR utilizes ultrasound waves in marine and underwater applications for navigation, underwater mapping, and locating objects like submarines and underwater structures.

4. Echocardiography:

In medicine, echocardiography uses ultrasound to create real-time images of the heart's structure and function, aiding in diagnosing heart conditions and monitoring cardiac health.

5. Ultrasonography:

Ultrasonography, commonly known as ultrasound, is a non-invasive medical imaging technique that uses high-frequency sound waves to visualize internal body structures in real-time. It is widely used for diagnosing and monitoring various medical conditions, including pregnancy, abdominal and pelvic disorders, and vascular diseases. The procedure involves the use of a handheld device called a transducer, which emits and receives sound waves to create detailed images of organs, tissues, and blood flow without the need for ionizing radiation.